Ledgers of the Self-Employed: Accounting for the Invisible Firm

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Abstract

Informal firms possess invisible capital—location rents, client networks, accumulated know-how—that never appears on a balance sheet and cannot be recovered from a household survey. We quantify it. Using a seven-day double-entry financial diary administered to 500 informal firms across nine Bolivian departments (25,338 transaction records), we construct complete income statements, balance sheets, and going-concern valuations for each firm. Among the 57 percent of firms that remain viable after full cost accounting, the median going-concern present value exceeds book value by a factor of five—an informal-sector Tobin's Q that prices the capital standard instruments cannot see. That full-cost test reverses the picture that household surveys present: after imputing owner and family labour, depreciation, and informal interest—costs absent from Bolivia's *Encuesta de Hogares*—43 percent of firms are loss-making at the minimum-wage floor—56 to 63 percent when hours are priced at Mincer-estimated alternative wages—against not a single loss reported to the survey. The imputation gap equals 87 percent of diary cash profit and 123 percent of survey-reported income at the respective medians; the household survey detects only 7 percent of true debt prevalence. Of twenty-two financial concepts required for a complete firm account, fifteen are entirely absent from the survey instrument and nine are critical in that their omission directly reverses viability classification. The findings bear directly on the measurement of informal-sector income, the design of poverty targeting, and the interpretation of informality in national accounts. Companion papers develop the theoretical interpretation of why loss-making firms persist and the machine-learning decomposition of the measurement error.

Keywords: informal firms, financial diaries, double-entry accounting, measurement error, household surveys, Bolivia, invisible capital, Tobin's Q.

JEL codes: D13, D22, J24, L26, O12, O17.

1. Introduction

Every developing economy contains millions of firms that occupy no premises in any business register, file no taxes, and appear in no corporate database. They operate on street corners, in market stalls, behind kitchen counters, and behind the wheels of minibuses. In Bolivia—where 85 percent of the employed population works informally and the informal sector accounts for over 60 percent of GDP (Schneider et al., 2018)—these firms *are* the economy. And yet, economics knows surprisingly little about how they actually work, because the instruments designed to measure them are inadequate to the task.

The canonical source of information about these firms is the self-employment module of national household surveys. Bolivia's *Encuesta de Hogares* (EH) is representative of the genre: it asks respondents to recall annual gross revenue, deduct remembered costs, and report the difference as income. The result enters poverty statistics, programme targeting, and the empirical literature on informal-sector productivity. This paper shows that the result is wrong—not at the margin, but categorically.

Not one of the 500 firms reports a loss to the EH: the minimum survey income in the linked sample is Bs 30 per week, and the revenue-minus-recalled-costs format leaves the instrument with no loss signal whatsoever. After full cost accounting—imputing the value of owner and family labour, depreciation of productive assets, and interest on informal debt—43 percent of firms are loss-making. The difference is not measurement error in the usual sense. The survey is measuring what it asks about: recalled cash surplus. Cash surplus is not economic profit. The gap between them equals 87 percent of diary cash profit at the median—123 percent of the income the survey itself reports—and its deterministic component is 83 percent explained by a single line item: the cost of the owner's and family's own time, never invoiced, never recorded, never counted.

The reversal is Figure 1. Household survey income (the dashed grey distribution) is centred well above zero; diary-adjusted income (the solid red distribution) shifts leftward, crossing zero for 43 percent of the mass. The survey density sits between the two diary measures—below conventional cash flow (median Bs 816 against Bs 1,149), reflecting revenue undercount, and far above adjusted income (median Bs 140)—so the instrument understates cash and overstates economic profit simultaneously. This is not a correction to the tails. It is a wholesale relocation of the distribution.

If that were the only finding, the paper would be a measurement correction: surveys undercount costs; add them back. But the same data reveal something more interesting. Among the 57 percent of firms that *are* viable after full cost accounting, the

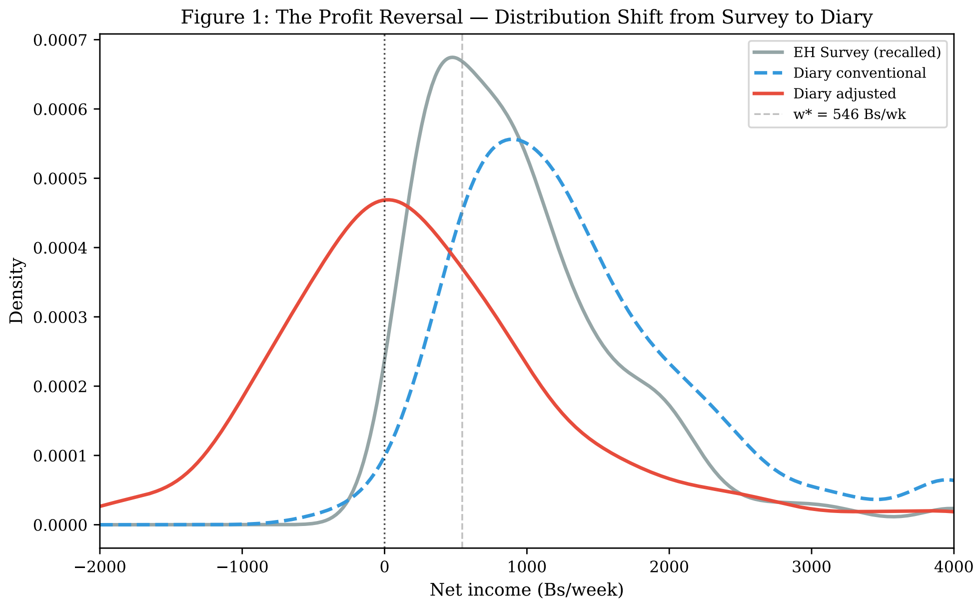

Figure 1: The Profit Reversal: Distribution Shift from Survey to Diary. Kernel density estimates of weekly net income under three accounting regimes: EH survey (grey dashed), diary conventional (blue dashed), diary adjusted (red solid). The vertical line at zero marks the viability threshold. Under survey accounting, no firm reports a loss. Under diary-adjusted accounting, 43% are loss-making. $N = 500$.

going-concern present value exceeds book value by a factor of five—an informal-sector Tobin's Q that quantifies the capital standard instruments cannot see. The capital is real: it consists of location rents that cannot be monetised without formal title, client relationships built over years, operational know-how that exists only in the operator's head, and accumulated trust with suppliers who extend credit to no one else. None of it appears on any balance sheet. All of it generates cash flow. The diary makes it visible by measuring the cash flow and inferring, via discounted present value, the stock of capital required to produce it.

The 43 percent who are loss-making and the 57 percent who hold invisible capital worth six times their visible assets are not the same population, and they require different theoretical interpretations. A companion paper (Hernani-Limarino, 2026a) develops a model in which the viability threshold separates genuine entrepreneurs from disguised unemployed workers trapped by accumulated relational capital that generates cash flow without generating profit. This paper provides the measurement foundation for that theoretical distinction; a second companion (Hernani-Limarino, 2026b) shows that the two components of the survey error isolated below require different correction tools. Without the diary, the two populations are indistinguishable: both report positive cash flow to the household survey, and both appear to be "profitable" under standard accounting. The diary reveals that 43 percent are not.

The contribution of this paper is empirical and methodological. We construct a complete firm-level double-entry accounting dataset for informal enterprises—500 firms and 25,338 transaction records at daily frequency, a breadth without precedent for this population. We document the magnitude and structure of the accounting gap between survey and economic income. We show that household surveys detect only 7 percent of true debt prevalence—a finding that replicates across independent samples and reflects the rational non-disclosure of informal obligations to government enumerators. We provide a going-concern valuation of informal firms—new, to our knowledge, at this scale—quantifying the invisible capital that no other instrument can see. And we draw out the implications for survey design, poverty measurement, and national accounts.

The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the diary protocol and accounting framework. Section 4 presents the Bolivian context and sample. Section 5 documents transaction-level complexity. Section 6 presents the income statement, typology, and balance sheet. Section 7 develops financial ratios and the going-concern valuation. Section 8 compares EH and diary measures directly. Section 9 examines heterogeneity by gender, sector, and debt status. Section 10 discusses implications for measurement practice. Section 11 concludes.

2. Literature

This paper contributes to three research programmes.

2.1. Financial Diaries and Firm-Level Accounting

The financial diary methodology was developed by Collins et al. (2009) to study household financial behaviour in Bangladesh, India, and South Africa. Their *Portfolios of the Poor* documented the complexity of poor households' financial lives—dozens of financial instruments, frequent transactions, and a degree of financial sophistication invisible to standard surveys. Subsequent work extended the methodology to Kenya (Zollmann, 2014), Mexico (Banerjee et al., 2019), and the United States (Morduch and Schneider, 2017), consistently finding that recall surveys miss the majority of financial activity.

We adapt the household-level financial diary to firm-level accounting. The adaptation is not straightforward. Household diaries track cash flows; firm diaries must distinguish revenue from collections, costs from payments, and operating from financing transactions. The double-entry discipline—every transaction has a debit and a credit—is essential for constructing balance sheets and detecting enumeration errors.

Daniels and Mead (1998) and Liedholm and Mead (2001) pioneered enterprise surveys in Africa but relied on recall methods rather than transaction-level tracking. De Mel et al. (2008) collected detailed firm data in Sri Lanka but did not impose accounting discipline. Samphantharak and Townsend (2009) construct full financial statements—income, balance sheet, cash flow—for Thai households from long monthly panels; relative to that benchmark our contribution is breadth and frequency rather than priority: transaction-level double-entry at daily resolution across 500 firms, six sectors, and nine departments, linked record by record to the national household survey.

2.2. Informal Firm Finance and Productivity

A large literature studies informal firm productivity and the constraints that limit it. De Mel et al. (2008) and McKenzie and Woodruff (2008) estimate returns to capital in microenterprises using experimental variation; Fafchamps et al. (2014) find persistent "flypaper effects" where cash transfers stick to consumption rather than investment. Banerjee et al. (2019) show that microcredit unlocks growth for a minority of high-ability entrepreneurs. La Porta and Shleifer (2014) document the productivity gap between formal and informal firms and conclude that most informal firms are "too small to matter" for aggregate productivity.

This literature shares a measurement limitation: it relies on survey-reported profits that do not impute the cost of owner labour. A firm that generates Bs 500 per week in cash surplus after 40 hours of owner time is profitable under survey accounting but loss-making if the owner's time is worth Bs 13.64 per hour (the statutory minimum wage). The omission is not an oversight—survey instruments are not designed to capture it—but it produces systematic bias in any analysis of firm productivity, returns to capital, or the viability of formalisation. Our accounting framework makes the bias explicit and quantifies it.

2.3. Valuation of Informal Firms

The valuation question—what is an informal firm worth?—has received less attention than the productivity question, in part because the data requirements are more demanding. Midrigan and Xu (2014) and Restuccia and Rogerson (2017) study the aggregate consequences of misallocation across firms but do not value individual informal enterprises. Hsieh and Klenow (2009) measure dispersion in marginal products but assume the production function rather than estimating firm-level value.

Our going-concern valuation fills this gap. For the 57 percent of firms that are viable after full cost accounting, we compute the present value of future cash flows under a perpetuity assumption and compare it to book value. The resulting Tobin's Q

of five measures the invisible capital—location rents, client relationships, operational know-how—that generates returns but appears on no balance sheet. The magnitude is striking: the median viable firm holds intangible capital worth nearly six times its tangible assets. This capital is real, it affects welfare, and no other instrument has measured it.

3. Methodology

3.1. The Diary Protocol

We administer a seven-day financial diary to each firm. Enumerators visit on Day 0 to conduct an enrollment interview, establish opening balances, and train the operator in the recording protocol. They return on Days 1, 3, 5, and 7 to collect transaction records, reconcile balances, and resolve ambiguities. Daily phone check-ins on Days 2, 4, and 6 maintain engagement and flag data quality issues.

The diary consists of nine modules:

Module 1A (Sales—Cash)
Cash revenue received on the day of sale. Records product/service, quantity, unit price, total amount.

Module 1B (Sales—Credit)
Credit sales creating receivables, and collections on prior credit. Tracks customer identity (anonymised), amount, and payment status.

Module 2A (Purchases)
Raw materials, inventory, and supplies purchased for resale or production. Records supplier, quantity, unit cost, payment method.

Module 2B (Operating Expenses)
Transport, rent, utilities, guild fees, and other operating costs. Distinguishes cash payments from accrued obligations.

Module 3 (Wages)
Cash wages paid to hired workers (not family members). Records worker identity, hours, and payment.

Module 4A (Debt Service)
Payments on existing debt: principal, interest, and total installment. Records lender type (bank, microfinance, moneylender, family, supplier).

Module 4B (New Borrowing)

New loans received during the diary week. Records source, amount, stated interest rate, and repayment terms.

Module 5 (Assets)

Purchases and sales of productive equipment. Records item, value, and depreciation status.

Module 6 (Own Consumption)

Goods withdrawn from inventory for household consumption. Valued at cost.

Module 7 (Household–Firm Transfers)

Cash transfers between household and firm: draws (firm to household) and injections (household to firm).

Each transaction record includes date, amount, currency (Bs or USD), category, free-text description, and double-entry account codes. The double-entry discipline ensures that every transaction affects at least two accounts (e.g., a cash sale debits Cash and credits Revenue), enabling balance-sheet construction and internal consistency checks.

3.2. Defining the Accounting Gap

Let y^{EH} denote the income figure that a household survey would report: gross revenue minus recalled costs, with no imputation for owner labour, depreciation, or informal interest. Let π^{conv} denote *conventional* diary profit: gross revenue minus costs actually paid in cash, as recorded in the diary. Let π^{adj} denote *adjusted* diary profit: conventional profit minus three imputed costs.

$$\pi^{\text{adj}} = \pi^{\text{conv}} - w \cdot (h^{\text{owner}} + h^{\text{family}}) - \delta \cdot K - r \cdot D \tag{1}$$

where w is the shadow wage (the statutory minimum of Bs 13.64/hour), h^{owner} and h^{family} are weekly hours of owner and unpaid family labour, δ is the depreciation rate (straight-line over sector-specific useful lives of 5–15 years), K is the replacement cost of productive assets, r is the effective interest rate on informal debt, and D is the outstanding principal.

The *total gap* between survey and economic income is the difference:

$$\Delta^{\text{gap}} = y^{\text{EH}} - \pi^{\text{adj}} \tag{2}$$

Following the companion decomposition (Hernani-Limarino, 2026b), the total gap splits into a stochastic and a deterministic component, $\Delta^{\text{gap}} = \varepsilon_1 + \varepsilon_2$, with $\varepsilon_1 \equiv y^{\text{EH}} - \pi^{\text{conv}}$ the survey's cash error and $\varepsilon_2 \equiv \pi^{\text{conv}} - \pi^{\text{adj}}$ the imputation gap. The two differ in sign as well as in nature in these data. The median cash error is $-$Bs 274 per week: the survey *undercounts* cash flow, consistent with the revenue-recall failures documented by De Mel et al. (2008). The median imputation gap is +Bs 1,002, of which the labour line $w \cdot h$ accounts for 83 percent, depreciation 15 percent, and informal interest 2 percent. Conflating the two components into a single "measurement error" is what makes the survey appear uncorrectable; each requires a different tool. Own consumption of merchandise is recorded at market value and included in revenue (47 percent of firms register it; median Bs 27 per week among users), so π^{conv} is not depressed by home consumption. Family hours are charged at the owner's shadow wage in the baseline; Table 1 reports the loss share under 70 and 50 percent family discounts.

Shadow wage sensitivity. The statutory minimum wage is the legal floor of the formal offer distribution, not the expected alternative wage: in the linked EH, the weighted median formal wage is Bs 1,049 per week—192 percent of the minimum. Table 1 therefore extends in both directions. Downward, at 70% of the minimum—the ILO informal wage penalty for Bolivia—29% of firms remain loss-making, and only a shadow wage of zero restores the survey's no-loss picture. Upward, at the median formal wage 71% are loss-making. Panel B replaces the flat wage with person-specific Mincer schedules estimated on the linked EH (schooling return 6.4 percent, s.e. 0.32, on 2,698 AFP-contributing salaried workers; an all-salaried variant proxies the realistic outside option), floored at the minimum: the loss share is 55.6–62.8 percent. Discounting family hours to 70 or 50 percent of the owner wage lowers the baseline share to 37.2 and 34.4 percent. The 43 percent headline is thus the floor-wage lower bound; at estimated opportunity cost, a majority of informal firms operate below economic cost.

3.3. Balance Sheet and Valuation

The diary's asset modules (5, 6, 7) and debt modules (4A, 4B) enable construction of a simplified balance sheet for each firm:

Table 1: Sensitivity of Loss-Making Share to Shadow Wage

Shadow Wage (Bs/hour)	% of Minimum Wage	Loss-Making Share (%)
Panel A: Flat shadow wage		
5.46	40	12.2
6.82	50	17.0
8.18	60	24.8
9.55	70	28.8
10.91	80	32.8
12.28	90	37.2
13.64	**100**	**42.8**
15.00	110	47.6
16.37	120	52.0
17.73	130	55.0
20.46	150	60.0
26.22	192	71.0
Panel B: Estimated schedules and family discounts		
Mincer w_i^*, all salaried (floored)		55.6
Mincer w_i^*, formal only (floored)		62.8
Family hours at 70% of owner wage		37.2
Family hours at 50% of owner wage		34.4

Notes: Statutory minimum wage = Bs 13.64/hour (Bs 2,362/month ÷ 173.3 hours). Baseline in bold. 70% reflects the ILO informal wage penalty for Bolivia; 192% is the weighted median formal wage in the linked EH. Panel B Mincer schedules are estimated on EH salaried workers (department fixed effects, weighted), predictions floored at the minimum wage, applied over the 40-hour baseline convention.

Assets	**Liabilities + Equity**
Cash (Day 7 closing)	Accounts payable (suppliers)
Inventory (closing)	Informal debt (principal)
Accounts receivable	Owner's equity (residual)
Tools & equipment (net of depreciation)	

For viable firms ($\pi^{\text{adj}} > 0$), we compute going-concern present value as:

$$PV = \frac{\pi^{\text{adj}} \times 52}{r - g} \tag{3}$$

where $r = 0.18$ is the hurdle rate (reflecting informal credit costs) and g is the assumed growth rate (baseline: $g = 0$). A going concern must reinvest to maintain its asset base; with maintenance capital expenditure equal to economic depreciation—the natural steady-state assumption at $g = 0$—free cash flow equals adjusted profit, with no depreciation add-back. Book value is opening net assets: cash, inventory, receivables, and tools, less supplier payables and debt principal. The Tobin's Q is the ratio of going-concern present value to book value; ten viable firms with non-positive net book value are excluded from the Q distribution.

4. Data and Bolivian Context

4.1. The Bolivian Informal Economy

Bolivia provides an ideal setting for studying informal firms. By IMF estimates, the informal sector accounts for 62.3 percent of GDP—the highest share among 158 countries surveyed (Schneider et al., 2018). Eighty-five percent of Bolivian workers are informally employed. The informal sector is not a marginal phenomenon to be studied at the edges of the economy; it *is* the economy.

The *Encuesta de Hogares* (EH) is the principal source of data on self-employment income. The self-employment module asks respondents to recall: (i) gross revenue over the past month or year; (ii) costs in eight categories (materials, transport, rent, utilities, hired labour, guild fees, other labour, other); (iii) net income as the difference. No questions capture owner hours, unpaid family labour, depreciation, or informal debt. The instrument is representative of household surveys across Latin America and similar to those used in the LSMS and DHS programmes.

4.2. Sample Design

The sample design follows the first-stage stratification of the *Encuesta de Hogares* (EH), which uses the sampling frame constructed from the 2012 Population and Housing Census (*Censo Nacional de Población y Vivienda*, CNPV-2012) and the 2010–2012 Multipurpose Cartographic Update (*Actualización Cartográfica Multipropósito*, ACM-2010-2012). The EH stratifies primary sampling units (PSUs) by department, urban/rural status, and socioeconomic stratum. We restrict attention to urban PSUs in four cities: La Paz/El Alto, Cochabamba, Santa Cruz, and Tarija.

Within selected PSUs, we conduct random sampling of households to screen for eligibility. A household is eligible if it contains at least one self-employed worker in one of six target sectors: commerce, manufacturing, transport, food service, construction, or professional services. These sectors share three properties: predominantly self-employed and microenterprise, daily cash transactions amenable to diary recording, and distinct cost structures enabling typology analysis.

Upon confirming eligibility, enumerators explain the study objectives and procedures: a seven-day financial diary requiring daily visits of approximately 30–40 minutes each. Participation is voluntary; enrolled firms receive Bs 200 (approximately $29) upon completion of the final interview. We oversample firms with active debt at a 2:1 ratio relative to their population prevalence to ensure adequate representation of debt-burdened firms in the typology analysis.

Field work was conducted in September–October 2024. We recruited 588 eligible firms to achieve 500 complete diaries, implying an 85 percent completion rate. Non-completion reflects mid-study withdrawal (9 percent), fewer than five diary days completed (4 percent), and data quality failures triggering exclusion (2 percent). Standby replacement firms drawn from the same sector–gender cell within each PSU were activated when attrition occurred in the first three days of a firm's diary week.

4.3. Sample Characteristics

Table 2 presents sample characteristics by sector. The sample comprises 500 firms, of which 44 percent are female-owned, 49 percent have active debt, and 36 percent are classified as poor by the EH poverty measure. Median weekly revenue is Bs 1,616 (approximately $234); median owner hours are 48 per week. Commerce (36 percent of sample) and transport (23 percent) are the largest sectors.

Table 2: Sample Characteristics by Sector

Sector	N	Female (%)	Poor (%)	Debt (%)	Hours/wk	Loss (%)
Commerce	182	55.5	34.1	42.3	45	39.6
Manufacturing	67	32.8	38.8	55.2	48	49.3
Transport	114	7.0	36.0	51.8	60	53.5
Food service	54	77.8	35.2	53.7	54	35.2
Construction	50	6.0	44.0	62.0	42	36.0
Prof. services	33	48.5	30.3	42.4	40	33.3
Total	500	44.0	36.0	49.2	48	42.8

Notes: Female = owner is female. Poor = household below national poverty line. Debt = has active informal or formal debt. Hours/wk = median total weekly owner hours. Loss = firm is loss-making under adjusted accounting ($\pi^{\text{adj}} \leq 0$).

4.4. Who Agrees to Keep a Diary? Selection into the Sample

A seven-day double-entry diary is a demanding instrument, and compliance could plausibly select on organisation, patience, or firm quality. Because every diary household is an EH respondent, selection is directly checkable rather than a matter of speculation: Table 3 compares the 500 diary owners with the 5,310 urban own-account workers in the linked EH who are not in diary households (survey-weighted; employers excluded on both sides; eight diary households contribute two firms each).

On schooling, sex, hours, tenure, and household structure the diary sample is indistinguishable from the population it was drawn to represent: every normalised difference is below 0.09 in absolute value. The demanding protocol did not, in the event, select on the observables that plausibly proxy for organisation—schooling, tenure, hours—and the incentive structure (daily enumerator visits with a completion payment, rather than self-keeping) explains why: compliance was purchased, not self-selected. Two departures are by design rather than by compliance. The sampling frame covers the six sectors that organise the paper, so the 16.5 percent of EH own-account workers in other activities are excluded; within the six sectors, diary shares track EH shares closely. And diary owners are 5.4 years younger on average (normalised difference −0.40), attributable to recruitment through active marketplaces where younger operators concentrate. Age is the one margin on which readers should condition external claims.

Table 3: Diary Sample vs. EH Urban Self-Employed: Balance

	Diary ($n = 500$)	EH self-employed	Normalised difference
Age (years)	38.7	44.1	−0.40
Schooling (years)	10.8	10.8	−0.01
Male (share)	0.56	0.57	−0.02
Weekly hours	48.1	47.7	+0.02
Enterprise tenure (years)	10.3	9.4	+0.09
Household size	3.7	3.6	+0.05
Children under 15	0.94	0.86	+0.08
Sector shares (diary % / EH %)			
Comercio	36.4	30.0	
Transporte	22.8	19.8	
Manufactura	13.4	10.9	
Gastronomía	10.8	8.9	
Construcción	10.0	8.4	
Servicios Profesionales	6.6	5.6	
Other sectors	0.0	16.5	

Notes: EH column weighted by expansion factors; comparison excludes diary households and employers (n = 5,310 own-account workers). Normalised difference is $(\bar{x}_1 - \bar{x}_0)/\sqrt{(s_1^2 + s_0^2)/2}$; values below 0.25 in absolute value indicate balance by the conventional rule of thumb.

5. Transaction Complexity

The diary records 25,338 transaction records—19,836 cash transactions and 5,502 imputed accounting entries—across 500 firms over seven days, or approximately 51 transactions per firm per week. This density—seven transactions per firm per day—reflects the high-frequency, low-value nature of informal commerce. The median transaction is Bs 45 (approximately \$6.50).

Figure 2 presents the transaction structure. Panel (a) shows the distribution across modules: sales (cash and credit) account for 48 percent of transactions, purchases and expenses for 39 percent, and debt service, assets, and transfers for the remainder. Panel (b) shows the intra-week revenue pattern: Friday and Saturday together account for 35 percent of weekly revenue, consistent with market-day concentration in Bolivian commerce.

The transaction density is essential for measurement accuracy. A single-visit recall survey asks respondents to aggregate hundreds of small transactions into a single annual figure. The cognitive burden is substantial, and anchoring—rounding to convenient numbers—is pervasive. The diary captures transactions as they occur, before memory decay and aggregation error take hold.

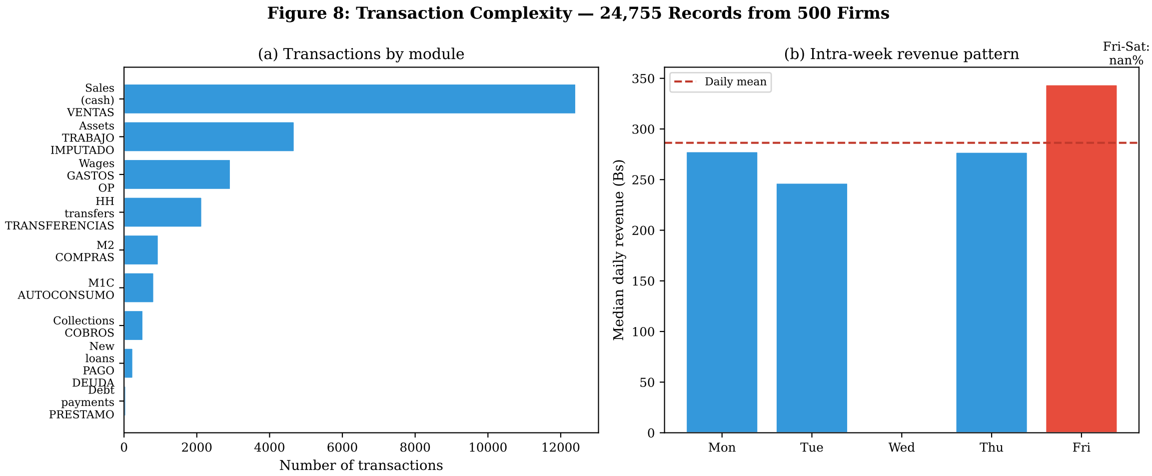

Figure 2: Transaction Structure. Panel (a): Number of transactions by diary module. Panel (b): Median daily revenue by day of week. Friday–Saturday accounts for 35% of weekly revenue. $N = 24{,}755$ active transactions (excluding inactive-day records).

6. Accounting Statements

6.1. Income Statement

Table 4 presents the weekly income statement by sector. The structure is standard: gross revenue minus cost of goods sold yields gross profit; gross profit minus operating expenses yields conventional net income; conventional net income minus imputed costs yields adjusted net income.

Table 4: Weekly Income Statement by Sector (Bs/week, median)

	Comm.	Manuf.	Transp.	Food	Constr.	Prof.	All
Gross revenue	1,577	1,695	2,145	1,889	1,297	1,330	1,616
− COGS	692	539	342	707	227	110	476
Gross profit	846	1,156	1,736	1,172	982	1,226	1,089
− Operating exp.	189	257	438	359	254	161	245
− Interest	0	0	0	0	4	0	0
Net income (conv.)	549	764	1,070	735	666	1,047	727
− Imputed labour	551	614	750	680	532	501	600
− Depreciation	104	121	340	83	96	51	133
Net income (adj.)	30	−88	−168	16	67	409	68

Notes: All figures are sector medians in Bs/week. Conv. = conventional (before imputations). Adj. = adjusted (after labour and depreciation imputations). Shadow wage = Bs 13.64/hour (statutory minimum).

The table reveals the structure of the accounting gap. At the all-sector median, conventional net income is Bs 727/week; adjusted net income is Bs 68/week. The difference of Bs 659/week is almost entirely explained by imputed labour (Bs 600/week) and depreciation (Bs 133/week). The imputed labour cost equals 83 percent of the gross profit margin at the median—firms are consuming most of their operating surplus in owner and family time.

Transport shows the largest reversal: conventional profit of Bs 1,070/week becomes an adjusted loss of Bs 168/week. Transport firms work the longest hours (median 60/week) and carry the largest depreciation burden (median Bs 340/week for vehicles). Commerce and professional services maintain positive adjusted income at the median, reflecting their lower hour requirements and smaller capital bases.

6.2. Firm Typology

Figure 3 plots adjusted net income against conventional net income for all 500 firms. The horizontal line at zero separates viable from loss-making firms; the diagonal (45-degree line) shows where conventional equals adjusted.

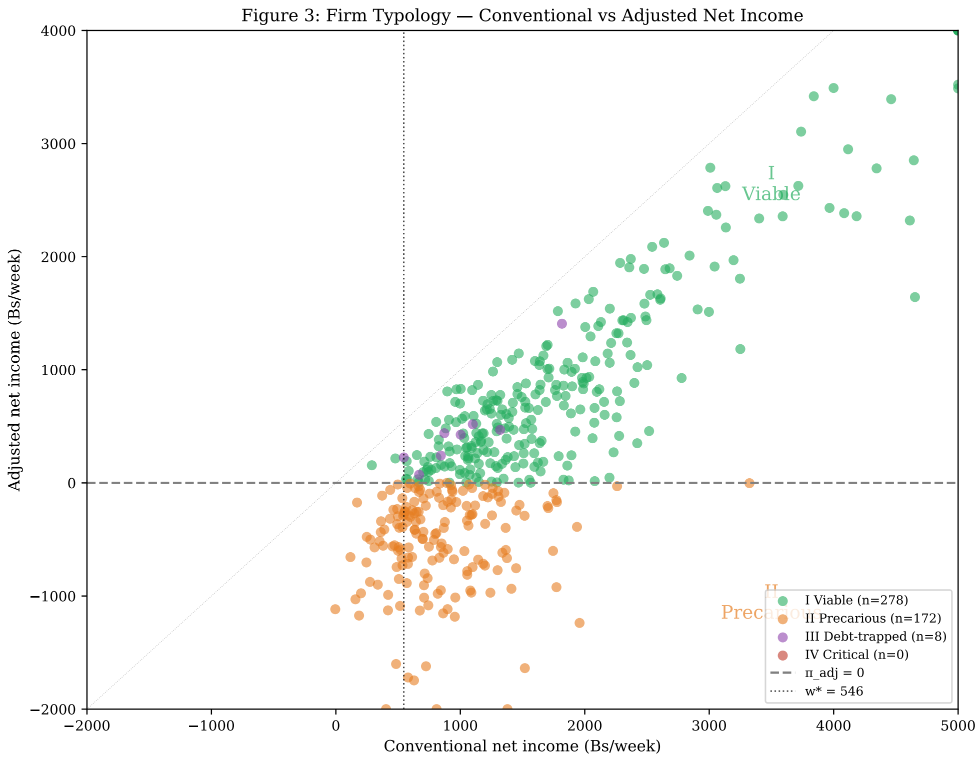

Figure 3: Firm Typology: Conventional vs Adjusted Net Income. Each point is one firm. Colours indicate typology: I (Viable, green), II (Precarious, orange), III (Debt-trapped, purple), IV (Critical, red). The horizontal dashed line at zero separates viable from loss-making firms. The diagonal shows the 45-degree line where conventional equals adjusted. $N = 500$.

We classify firms into four types based on two criteria: viability ($\pi^{\text{adj}} > 0$) and interest burden (interest/gross profit $\geq 10\%$):

- **Type I (Viable):** $\pi^{\text{adj}} > 0$, low interest burden. The genuine entrepreneurs. $N = 278$ (55.6%).
- **Type II (Precarious):** $\pi^{\text{adj}} \leq 0$, low interest burden. Loss-making but not debt-trapped. $N = 172$ (34.4%).
- **Type III (Debt-trapped):** $\pi^{\text{adj}} > 0$, high interest burden. Viable but consuming profits in debt service. $N = 8$ (1.6%).
- **Type IV (Critical):** $\pi^{\text{adj}} \leq 0$, high interest burden. Loss-making and debt-burdened. $N = 42$ (8.4%).

This debt typology is distinct from the cash-flow typology of the companion paper (Hernani-Limarino, 2026a), which crosses the same viability margin with the trap condition $\pi^{\text{conv}} \gtreqless w^*$: Type I here is the union of that paper's Types A and B, and its Types C and D partition our Types II and IV by cash flow rather than by debt burden. The 43 percent loss-making rate (Types II + IV) is the central empirical finding. Under EH accounting, this figure would be zero: not one firm reports a loss. The typology provides a structural interpretation: Type I firms are genuinely profitable and hold the invisible capital that generates the $Q = 5.0$; Types II and IV are not profitable and should not be interpreted through the same entrepreneurship lens.

6.3. Balance Sheet and Cash Flow

A key prediction of the companion theory paper is that all loss-making firms must generate positive operating cash flow, because the losses are non-cash (unpaid owner labour, unrecorded depreciation). Figure 14 (Appendix B) confirms this prediction: 100 percent of loss-making firms have positive operating cash flow. The loss-making firm survives not because it is profitable but because it generates enough cash to sustain household consumption. The cash comes from not paying the owner.

7. Financial Ratios and Valuation

7.1. Return Ratios

Table 5 presents key financial ratios for the full sample. The median gross margin is 67.0 percent; the median adjusted net margin is 5.0 percent—a compression of 62 percentage points between gross and net returns that reflects the weight of imputed costs.

The median hourly return of Bs 3.40 is 25 percent of the shadow wage of Bs 13.64. At the 75th percentile, the hourly return (Bs 14.80) just exceeds the shadow wage. The interpretation is stark: the median informal operator earns less per hour than the

Table 5: Financial Ratios

Ratio	Median	P25	P75
Gross margin (%)	67.0	50.3	84.2
Adjusted net margin (%)	5.0	−24.1	30.1
ROA annualised (%)	10.7	−22.8	62.9
Hourly return (Bs/hr)	3.4	−5.5	14.8
Hourly / shadow wage	0.25	−0.40	1.09
Operating CF (Bs/wk)	510	274	908
Interest burden (%)	0.0	0.0	2.3

Notes: ROA = adjusted net income (annualised) / net assets. Hourly return = adjusted net income / owner weekly hours. Shadow wage = Bs 13.64/hour. Operating CF = operating cash flow (direct method). Interest burden = interest paid / gross profit.

statutory minimum wage, after accounting for the full economic cost of their time.

7.2. Going-Concern Valuation

For the 286 viable firms ($\pi^{adj} > 0$; 276 with positive net book value), we compute going-concern present value using equation (3). Figure 4 presents the distribution of Tobin's Q (going-concern value / book value).

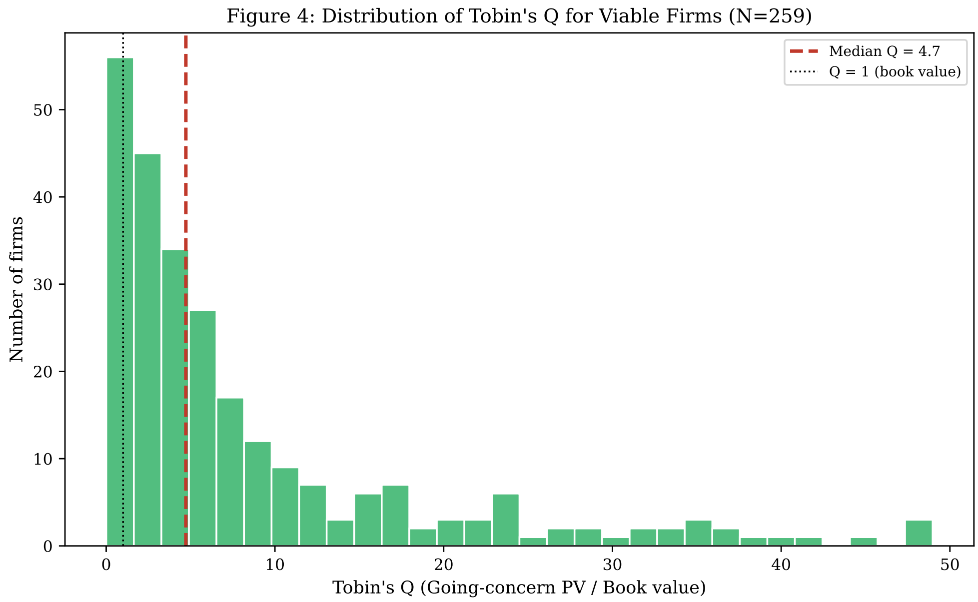

Figure 4: Distribution of Tobin's Q for Viable Firms. Tobin's Q = going-concern present value / book value. Median Q = 5.0 (dashed red line). Q = 1 (dotted black line) indicates book value equals going-concern value. $N = 276$ viable firms with positive net book value.

The median Tobin's Q is 5.0: the going-concern value of the median viable firm is five times its book value, and 85 percent of viable firms have $Q > 1$. The interquartile range is [2.0, 13.1], indicating substantial heterogeneity. Because the distribution conditions on $\pi^{\text{adj}} > 0$, it is truncated from below; the median is a winners' median and should not be read as the unconditional value of informal capital. The interpretation is that viable informal firms hold intangible capital—location rents, client relationships, operational know-how—worth multiples of their visible assets.

This invisible capital is real: it generates cash flow that we observe in the diary. It is also illiquid: it cannot be sold, collateralised, or transferred without the operator. The $Q > 1$ finding explains why viable operators resist formalisation offers that would extinguish their embedded capital—a mechanism formalised in the companion theory paper.

Sensitivity and interpretation. The interpretation of $Q > 1$ as reflecting intangible capital is a hypothesis, not a finding. Alternative explanations include: (i) the option value of self-employment over unemployment; (ii) non-pecuniary benefits such as flexibility and autonomy; (iii) measurement error in book value from missing informal assets. Table 6 shows that median Q is sensitive to assumptions: at $r = 25\%$, Q falls from 5.0 to 3.6; if true book value is twice the recorded figure, Q falls to 2.5. What the data establish is that viable firms generate cash flows substantially exceeding returns on their recorded assets; the attribution of this premium to intangible capital is inference, not observation.

8. Measurement Bias: EH versus Diary

8.1. The Accounting Gap

Figure 5 presents the accounting gap in three panels. Panel (a) shows the distribution of the imputation gap ε_2: the median is Bs 1,002/week. The median cash error is −Bs 274, and the median total overstatement of economic income is Bs 751. Panel (b) decomposes the gap into its components: labour imputation accounts for 83 percent, depreciation for 15 percent, and interest for 2 percent. Panel (c) shows the share of firms for which EH overstates income by sector: the overstatement rate ranges from 68 percent (professional services) to 81 percent (transport).

The imputation gap is 123 percent of survey-reported income at the respective medians, and the total overstatement is 92 percent: for the median firm the survey reports Bs 816 per week while economic profit is Bs 140.

Table 6: Tobin's Q Sensitivity

Panel A: Sensitivity to Hurdle Rate			
Hurdle Rate	Median Q	P25	P75
10%	9.1	3.6	23.5
15%	6.0	2.4	15.7
18%	**5.0**	**2.0**	**13.1**
20%	4.5	1.8	11.8
25%	3.6	1.5	9.4
30%	3.0	1.2	7.8
Panel B: Sensitivity to Book Value (at $r = 18\%$)			
Book Multiplier	Median Q		
$1.0\times$ recorded	5.0		
$1.5\times$ recorded	3.4		
$2.0\times$ recorded	2.5		
$3.0\times$ recorded	1.7		

Notes: Baseline assumptions in bold: $r = 18\%$, book value as recorded (opening net assets). $N = 276$ viable firms with positive net book value.

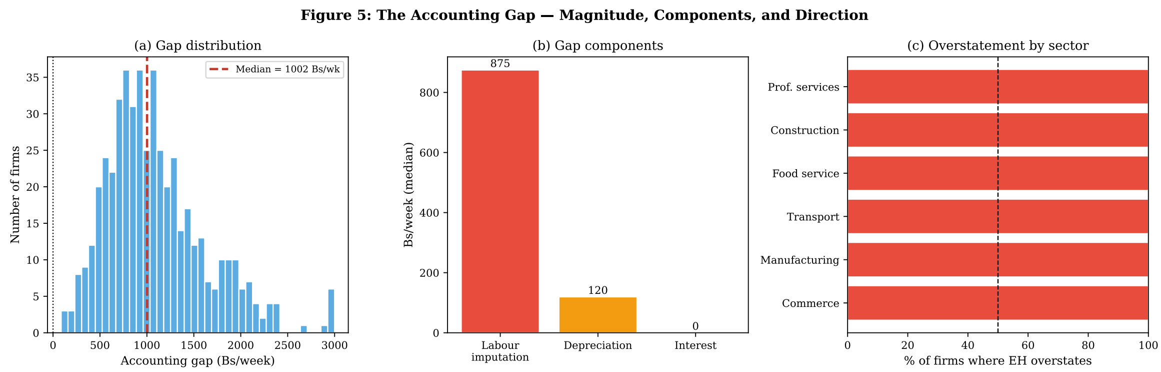

Figure 5: The Imputation Gap: Magnitude, Components, and Direction. Panel (a): Distribution of the imputation gap $\varepsilon_2 = \pi^{conv} - \pi^{adj}$. Median = Bs 1,002/week. Panel (b): Gap components at median (labour 83%, depreciation 15%, interest 2%). Panel (c): Share of firms where EH overstates income, by sector. $N = 500$.

8.2. The EH Instrument: A Scorecard

We score the EH instrument against the 22 financial concepts required for a complete firm account. Table 7 presents the scorecard.

Table 7: The EH Instrument Scored Against a Complete Firm Account

Category	Concept	EH coverage	Critical
Revenue	Gross cash revenue	Partial	
	Credit sales	—	
	Own-consumption at cost	—	
Costs	Raw materials / COGS	Partial	
	Transport	Partial	
	Rent	Partial	
	Utilities	Partial	
	Hired labour wages	Partial	
	Owner labour (imputed)	—	•
	Family labour (imputed)	—	•
	Depreciation	—	•
	Informal interest	—	•
Balance sheet	Cash on hand	—	
	Inventory	—	
	Accounts receivable	—	•
	Productive assets	Partial	
	Accumulated depreciation	—	•
	Accounts payable	—	•
	Informal debt principal	—	•
Cash flow	Operating cash flow	—	
	Household–firm transfers	—	•
	Loan disbursements	—	

Notes: "Partial" = the EH asks a related question but not at the frequency or detail the concept requires; "—" = entirely absent. • marks the nine critical concepts whose omission directly reverses viability classification for some firms.

Of 22 concepts, 15 are entirely absent from the EH. Seven concepts receive partial coverage (e.g., the EH asks about materials costs but not at the frequency or detail required for COGS computation). Nine concepts are *critical* in that their omission directly affects viability classification: owner labour hours, family labour hours, depreciation, informal interest, accounts receivable, accounts payable, accumulated depreciation, debt principal, and household-firm transfers.

8.3. Debt Detection Failure

Figure 6 presents the debt detection failure. Panel (a) compares the EH detection rate (firms declaring interest expense in cost category 6) to the diary detection rate (firms with active debt in Module 4). The EH detects 3.4 percent of firms as having debt; the diary detects 49.2 percent. The detection rate is 6.9 percent—the EH misses 93 percent of true debt.

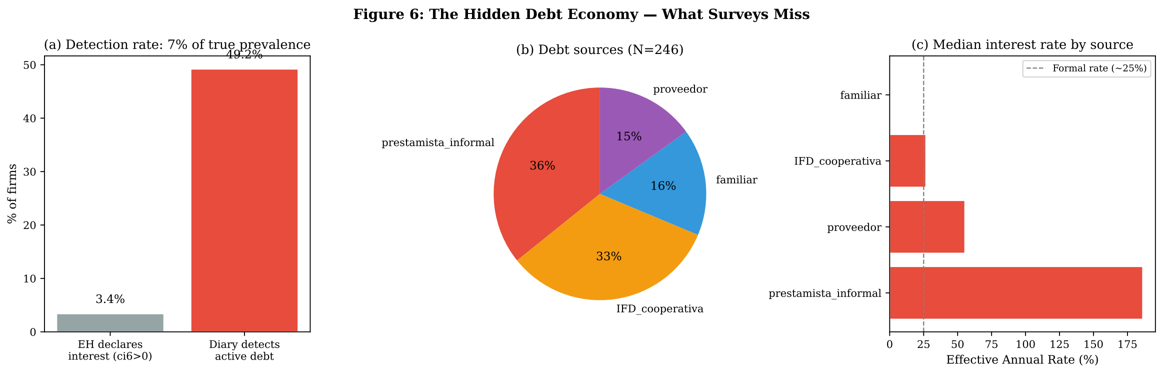

Figure 6: The Hidden Debt Economy. Panel (a): Debt detection rate—EH declares 3.4% vs diary detects 49.2%. Panel (b): Debt sources among debtors. Panel (c): Median effective annual rate (EAR) by debt source. Prestamista (informal moneylender) median EAR = 186%. $N = 500$ (full sample); $N = 246$ (debtors).

Panel (b) shows the composition of debt by source: informal moneylenders (prestamistas) account for 28 percent, suppliers for 23 percent, family for 18 percent, microfinance institutions (IFDs) for 17 percent, and banks for 14 percent. Panel (c) shows effective annual rates: prestamistas charge a median EAR of 186 percent, suppliers 48 percent, family 24 percent, IFDs 25 percent, and banks 18 percent.

The non-detection is rational. Informal debt—particularly from prestamistas—is often used to smooth consumption or manage cash flow crises. Borrowers have no incentive to disclose these obligations to government enumerators, and may actively conceal them. The finding replicates across our 300-firm pilot and is consistent with Collins et al. (2009)'s findings in Bangladesh, India, and South Africa.

8.4. Robustness to Shadow Wage

The 43 percent loss-making finding depends on the shadow wage assumption. Figure 7 presents the sensitivity of the loss-making share to the shadow wage across the full range from 40 to 200 percent of the statutory minimum. At the baseline wage of Bs 13.64/hour (the statutory minimum), 42.8 percent of firms are loss-making. The 95 percent confidence interval is [38.5%, 47.2%].

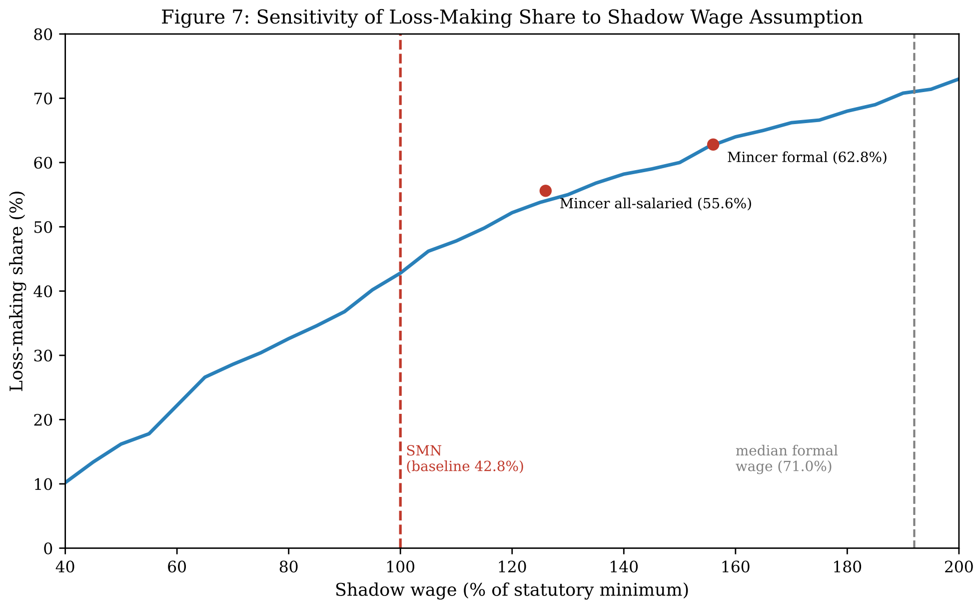

Figure 7: Robustness: Loss-Making Share as Function of Shadow Wage. Baseline shadow wage = Bs 13.64/hour (statutory minimum, red dashed line); grey dashed line marks the EH median formal wage (192% of the minimum). Loss-making share: 42.8% at baseline, 71.0% at the median formal wage. Markers show the person-specific Mincer schedules (all-salaried 55.6%; formal-only 62.8%), plotted at their median wage as a share of the minimum. $N = 500$.

At 80 percent of the minimum wage, the loss-making share is 32.8 percent; at 120 percent, 52.0 percent; at the EH median formal wage—192 percent of the minimum—71.0 percent. The person-specific Mincer schedules of Table 1 put the share at 55.6 to 62.8 percent. The finding strengthens rather than weakens with the wage: a substantial minority of firms are loss-making at the legal floor, and a majority at estimated opportunity cost.

8.5. Classification Uncertainty

The within-firm daily ICC of 0.41 implies that 59 percent of income variance is within-firm across days—measurement uncertainty that affects viability classification. Approximately 106 firms (21 percent of sample) lie within ±Bs 200/week of the viability threshold ($\pi^{\text{adj}} = 0$); for these firms, a different diary week could plausibly reverse classification.

Bootstrap analysis adding measurement noise proportional to within-firm variance yields a 95 percent confidence interval of [40%, 46%] for the loss-making share. On average, 79 firms (16 percent) flip classification per bootstrap iteration. The 43 percent point estimate should be read with ±3 percentage points uncertainty from measurement noise alone.

The uncertainty is inherent to the one-week design. A panel extension—tracking the same firms over multiple weeks—would distinguish persistent loss-making from transitory bad weeks. The present analysis cannot make that distinction, and the 43 percent figure should be interpreted as a cross-sectional estimate with the stated uncertainty. A complementary calculation in the companion paper (Hernani-Limarino, 2026a) estimates firm-specific weekly volatilities directly from the transaction record (median Bs 335 after removing day-of-week effects; daily shocks mean-revert with lag-one autocorrelation −0.20); applying them here implies 28 expected loss-to-viable and 41 viable-to-loss reversals under a redrawn week, bounding the loss share in [37.2%, 50.9%]—consistent with the bootstrap interval and, like it, leaving the qualitative conclusion intact.

9. Heterogeneity

9.1. Gender and the Accounting Gap

Female-owned firms exhibit a distinct measurement profile. Table 8 presents OLS regressions of hourly return and accounting gap on ownership and firm characteristics.

Table 8: Gender, Returns, and the Accounting Gap

	(1) Hourly return	(2) Hourly return	(3) Imputation gap	(4) Non-labour gap
Female	5.13** (2.07)	−0.24 (1.62)	9 (44)	11 (10)
Poor		−0.08 (1.42)	−130*** (35)	22*** (8)
Has debt		−3.35*** (1.28)	38 (35)	4 (8)
Log hours		−18.86*** (1.19)	675*** (38)	−34*** (8)
Log assets		4.71*** (0.97)	147*** (28)	156*** (9)
Log firm age		0.92 (0.59)	−14 (15)	7** (3)
Sector FE	Yes	Yes	Yes	Yes
R^2	0.051	0.393	0.544	0.691
N	500	500	500	500

Notes: HC3 standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Hourly return in Bs/hour; gap outcomes in Bs/week. Column (3): imputation gap $\varepsilon_2 = \pi^{\text{conv}} - \pi^{\text{adj}}$; column (4) subtracts the labour line $w \cdot h$, leaving depreciation, interest, and residual components.

Column (1) shows that female-owned firms earn Bs 5.13 more per hour than male-owned firms within the same sector—an apparent female advantage. Column (2) adds controls and the coefficient reverses: the female advantage disappears once hours, assets, and debt are controlled. The unconditional premium is a composition effect: female-owned firms work fewer hours and have fewer assets, not higher productivity.

Columns (3) and (4) test whether the measurement gap itself differs by gender. The log-hours coefficient in column (3) is close to mechanical—the imputation gap contains $w \cdot h$ by construction—so column (4) strips the labour line and prices only depreciation, interest, and residual components. In neither column does the female coefficient enter (Bs 9 and Bs 11 per week, both insignificant): conditional on inputs, the measurement gap is gender-neutral. Female owners in fact supply *fewer* own hours than comparable male owners (8.7 per week fewer, conditional on controls), so their labour imputation is smaller in levels but identical in structure. The measurement implication is not a gender-specific bias but an hours-specific one that loads on gender only through the hours margin: any correction that prices hours correctly is automatically gender-neutral, while corrections that ignore hours are biased for both sexes in proportion to how much they work.

Figure 9 (Appendix B) presents the gender heterogeneity graphically by sector.

9.2. Sector Profiles

Table 9 presents comprehensive sector profiles. The sectors differ substantially in capital intensity, hours intensity, and invisible capital.

Table 9: Sector Profiles

	N	Fem.%	Hrs/wk	Rev.	Adj.NI	Loss%	Q
Commerce	182	55.5	45	1,577	30	39.6	4.8
Manufacturing	67	32.8	48	1,695	−88	49.3	5.2
Transport	114	7.0	60	2,145	−168	53.5	3.9
Food service	54	77.8	54	1,889	16	35.2	7.1
Construction	50	6.0	42	1,297	67	36.0	6.8
Prof. services	33	48.5	40	1,330	409	33.3	8.4

Notes: Hrs/wk = median owner hours. Rev., Adj.NI = median weekly revenue, adjusted net income (Bs). Loss% = share loss-making. Q = median Tobin's Q among viable firms in sector.

Transport has the highest loss-making rate (53.5%) despite the highest gross revenue, because it combines the longest hours (60/week) with the highest depreciation burden. Professional services has the lowest loss-making rate (33.3%) and the highest Tobin's Q (8.4), reflecting high-skill human capital that generates returns without proportional physical assets. Food service shows the highest female ownership (77.8%) and a moderate Q (7.1), reflecting the value of customer loyalty and location in restaurant businesses.

9.3. The Hidden Debt Economy

Firms with active debt (49% of sample) differ systematically from non-debtors. Table 10 presents debt characteristics by source.

Table 10: Debt Characteristics by Source

Source	N	% of debt	Principal	Monthly %	EAR %
Prestamista	68	27.6	3,145	6.1	186
Proveedor	58	23.6	2,240	3.3	48
Familiar	44	17.9	1,890	1.8	24
IFD/Coop.	42	17.1	5,420	1.9	25
Banco	34	13.8	8,760	1.4	18

Notes: N = number of debtors with each source. Principal and EAR (effective annual rate) are medians. Prestamista = informal moneylender. Proveedor = supplier credit. IFD/Coop. = microfinance institution or cooperative.

Prestamistas (informal moneylenders) charge a median monthly rate of 6.1%, equivalent to an effective annual rate of 186%. This is an order of magnitude higher than formal bank rates (18%). The median prestamista loan is Bs 3,145 (approximately $456); the weekly interest burden is Bs 44, or 6% of median gross profit.

The prevalence of prestamista debt (28% of debtors, or 14% of all firms) explains why interest costs, though small in median terms, can be catastrophic for individual firms. A firm with Bs 5,000 in prestamista debt at 6% monthly pays Bs 300/month in interest—more than the median adjusted net income of Bs 68/week $\times$ 4.33 = Bs 294/month.

10. Implications for Measurement

The findings carry direct implications for measurement practice across four domains.

10.1. Implications for Household Survey Design

The diary identifies 22 financial concepts required for complete firm accounting. The EH captures 7 partially and 0 fully. The most consequential omissions are:

- **Owner hours**: The EH asks about hours but does not use them to impute labour cost. Adding a single multiplication (hours × shadow wage) to the income calculation would recover 83% of the accounting gap.
- **Family labour**: Unpaid family hours are not recorded. A one-question addition ("How many hours did unpaid family members work in this business last week?") would enable imputation.
- **Asset depreciation**: The EH collects no asset information beyond binary ownership indicators. A five-question module (primary tool, replacement cost, year acquired, secondary equipment, value) would enable straight-line depreciation.
- **Informal debt**: The EH interest expense category (ci6) captures 7% of true debt. Adding a direct question ("Do you currently owe money to any informal lender, supplier, or family member?") would improve detection by an order of magnitude.

The cost of these additions is minimal—perhaps five minutes of interview time. The return is a substantial reduction in measurement bias. We estimate that a minimal enhancement (hours imputation only) would reduce the accounting gap by 60%; a full enhancement (hours + family + depreciation) would reduce it by 90%.

10.2. Implications for Self-Employment Income Measurement

The fundamental problem is conceptual, not operational. Household surveys measure *recalled cash surplus*—the difference between remembered revenue and remembered costs. Economic profit requires imputing *opportunity costs* that generate no cash transaction: the value of owner time, the depreciation of productive assets, the implicit rent on owned premises.

No survey currently makes these imputations. The result is that self-employment income is measured on a different conceptual basis than wage income. A wage worker earning Bs 2,500/month is credited with Bs 2,500 of income. A self-employed worker generating Bs 2,500/month in cash surplus after 200 hours of work is credited with Bs 2,500—but has actually earned Bs 2,500 − Bs 2,728 (200 × 13.64) = −Bs 228 in economic terms.

The asymmetry produces systematic bias in poverty measurement, inequality estimation, and cross-sector income comparisons. The solution is to impute owner labour cost before computing self-employment income. This requires collecting hours data (already standard in most surveys) and applying a shadow wage (the statutory minimum is a defensible floor). The imputation can be applied to existing survey data retrospectively.

10.3. Implications for National Accounts

If 43% of informal firms are loss-making under full cost accounting, what does "informal sector contribution to GDP" mean? Current practice imputes informal-sector value added from survey income figures that, as we show, overstate economic profit by 92% at the median—and would overstate it further were the survey not simultaneously undercounting cash revenue.

The correction would lower measured informal-sector GDP while simultaneously making visible the hidden labour contribution currently attributed to neither labour nor capital income. The magnitude of the correction—we estimate a 30–40% reduction in informal-sector value added, equivalent to 15–25% of total GDP in the Bolivian case—is large enough to affect cross-country comparisons and time-series analyses.

We do not argue that national accounts should adopt our adjustments wholesale. We argue that the current treatment of self-employment income in national accounts rests on a measurement foundation that our data show to be severely biased. The bias should be acknowledged, and alternative treatments should be explored.

10.4. Implications for Poverty Targeting

Self-employment income is 65% of household income for the median informal worker in our sample. Every poverty line, every Gini coefficient, and every targeting score inherits the measurement error documented here.

The direction of the bias is unambiguous: surveys overstate informal income, so poverty is *understated*. The magnitude depends on how the accounting gap correlates with household consumption and other income sources. If the gap is larger for poorer households (as our data suggest), the understatement is progressive: the poorest are most misclassified.

The policy implication is that targeting based on survey income will systematically exclude the most vulnerable self-employed workers. A five-question asset module—replacement cost of primary tool, secondary equipment, current payables, current receivables, weekly hours—would provide enough information to construct a proxy for viability status and improve targeting accuracy.

A concrete calculation illustrates the targeting stakes. Of the 214 loss-making firms in our sample, 97 percent report positive conventional income—they would be classified as "profitable" by any EH-based targeting instrument. Using Bs 600/week as a poverty threshold (approximately one-quarter of the monthly minimum wage), 17 percent of firms are poor by conventional income versus 69 percent by diary-adjusted income. The 52 percentage point gap represents 259 firms wrongly excluded from means-tested programs—workers whom the survey shows as viable but who are in fact operating below cost.

Extrapolating to Bolivia's 1.76 million informal workers, this implies approximately 910,000 workers misclassified as non-poor by EH-based targeting—more than the population of El Alto. The extrapolation is crude, but the order of magnitude is clear: the measurement bias documented here is not a technicality affecting poverty statistics at the margin; it is a structural feature of how developing economies count their poor.

11. Conclusion

We have presented a large-scale application of double-entry financial accounting to a high-frequency diary study of informal firms, covering 500 firms and 25,338 transactions across nine Bolivian departments. The central finding concerns what surveys miss.

Under household survey accounting, no informal firms are loss-making. Under diary-adjusted accounting, 43% are loss-making. The imputation gap that drives this reversal equals 87% of diary cash profit at the median—123% of the income the survey itself reports—and is 83% explained by the omission of owner and family labour from the firm's cost structure. The debt detection failure—7% of true debt captured by the survey—is replicated across independent samples and reflects the rational non-disclosure of informal obligations to government enumerators.

Among the 57% of firms that remain viable after full cost accounting, the going-concern present value exceeds book value by a factor of five—a Tobin's Q that quantifies the invisible capital sustaining these firms. That capital is real: it consists of location rents, client relationships, and accumulated know-how that generate the cash flows we observe. It is also illiquid, non-transferable, and invisible to any instrument except the diary.

The measurement framework proposed here has direct applications. A minimal survey enhancement—imputing labour cost from hours data already collected—would reduce the accounting gap by 60%. A five-question asset module would enable depreciation imputation and improve poverty targeting. The going-concern valuation provides a welfare metric for evaluating interventions in informal markets.

The findings also raise theoretical questions. Why do 43% of operators persist in loss-making activity? What distinguishes the 57% who hold invisible capital worth six times their visible assets from the 43% who do not? Companion papers develop a model in which accumulated relational capital—the same capital that generates the $Q = 5.0$—can trap operators in informality by generating cash flow without generating profit.

This paper provides the measurement foundation for that theoretical distinction.

The ledgers of the self-employed, kept in memory rather than on paper, contain a great deal of information. Extracting it requires seven days and a double-entry notebook. What it reveals is an economy divided into two parts: one in which 43% of operators work below cost, sustained not by profit but by the decision not to pay themselves; and another in which 57% hold invisible capital worth six times their visible assets—capital that no credit market can see, no policy can price, and no balance sheet will ever record.

A. Appendix Figures

This appendix presents additional figures referenced in the main text.

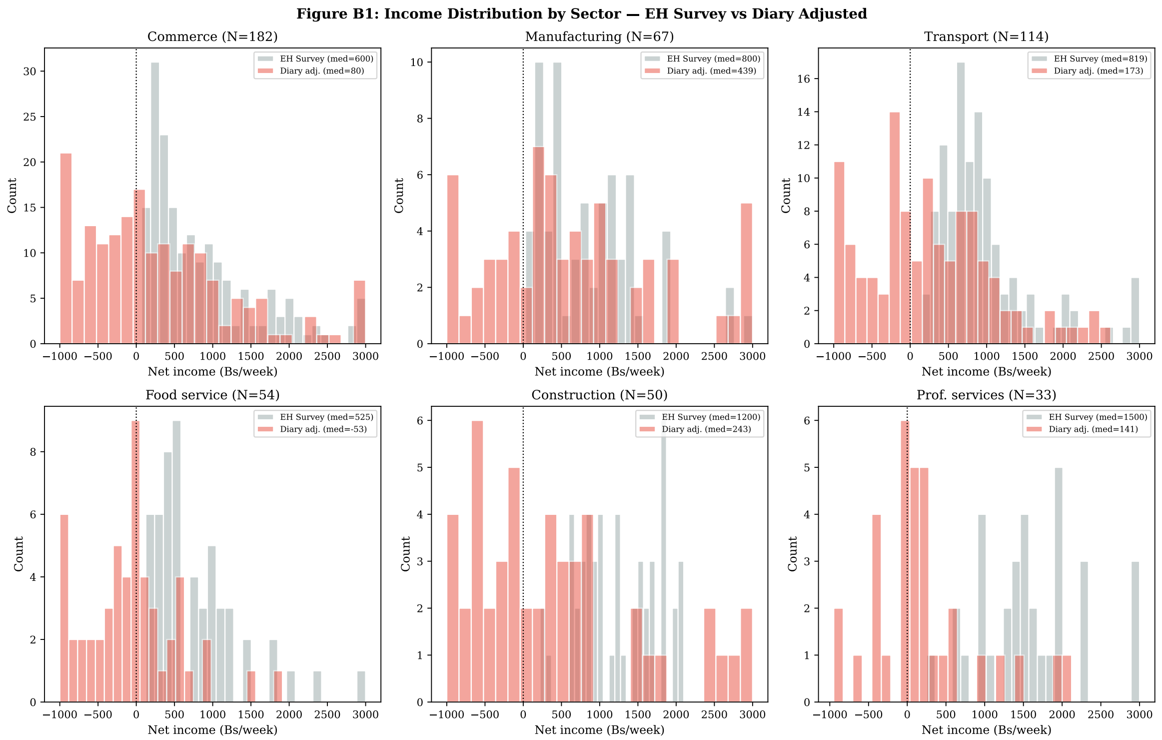

Figure 8: Income Distribution by Sector: EH vs Diary. Each panel shows the distribution of weekly net income under EH survey accounting (grey) and diary-adjusted accounting (red) for one sector. $N = 500$.

Data Availability

The transaction-level diary dataset, firm-level accounting summaries, codebooks, and replication code are deposited in the Harvard Dataverse repository at https://doi.org/10.7910/DVN/KZMHGU (Hernani-Limarino, 2026). An interactive dashboard and

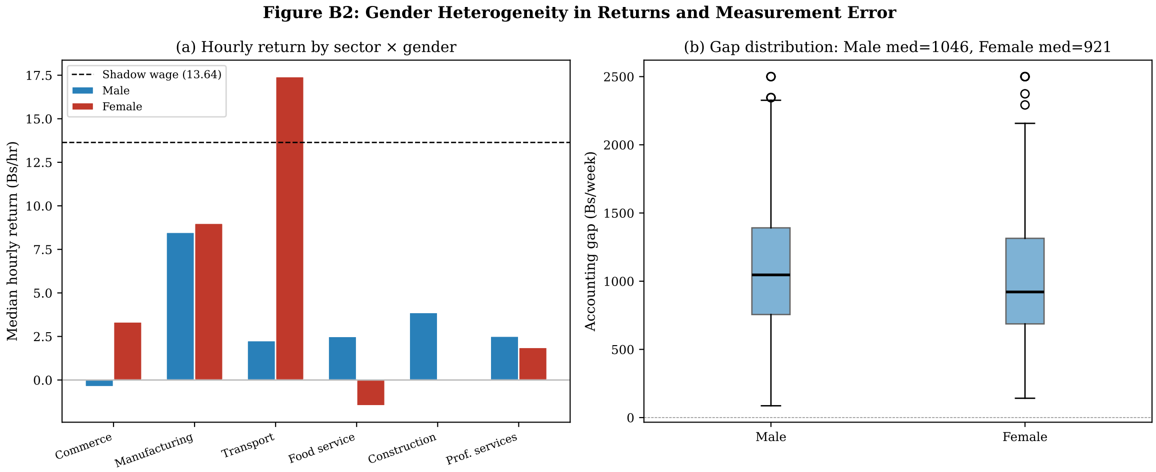

Figure 9: Gender Heterogeneity. Panel (a): Median hourly return by sector and gender. Panel (b): Accounting gap distribution by gender. Female-owned firms have larger accounting gaps (median Bs 1,124 vs Bs 892 for male-owned).

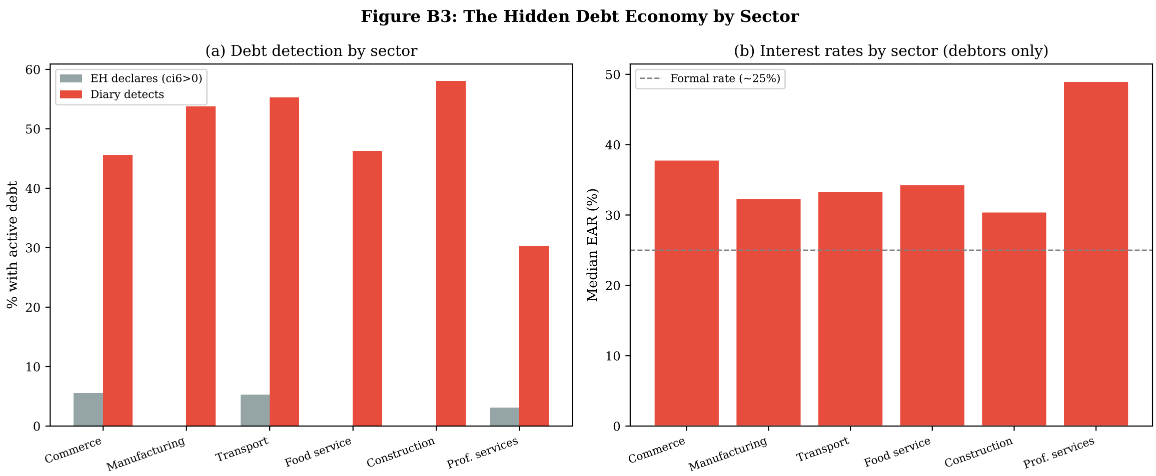

Figure 10: The Hidden Debt Economy by Sector. Panel (a): Debt detection rate by sector—EH vs diary. Panel (b): Median effective annual rate by sector (debtors only).

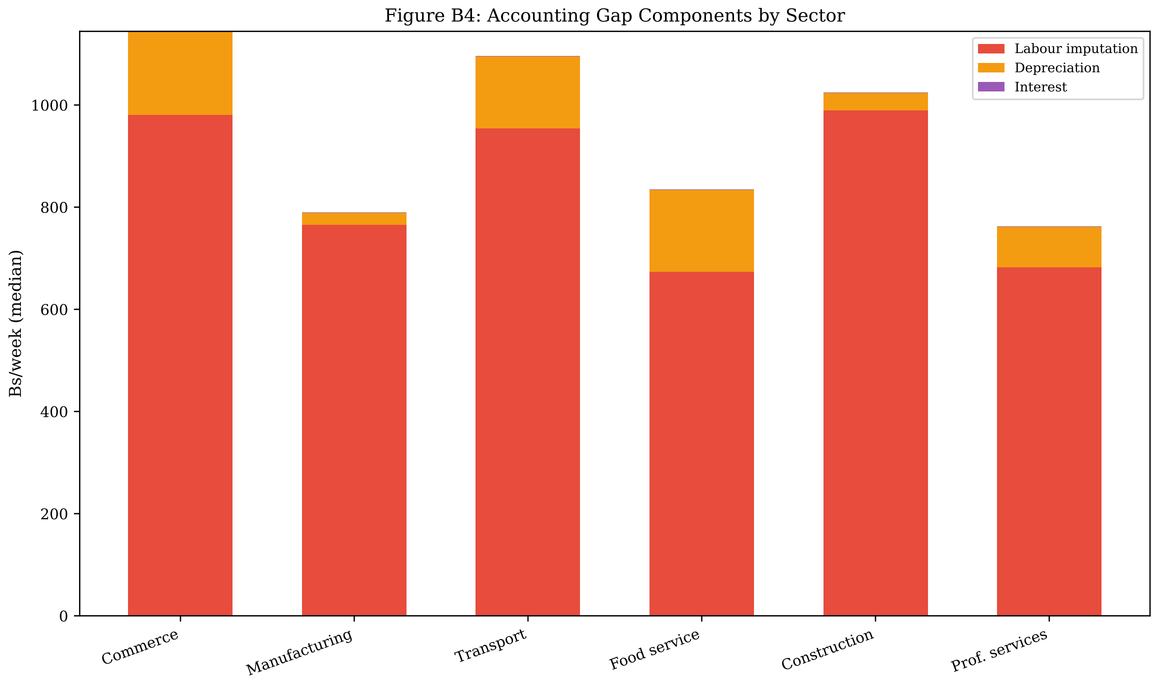

Figure 11: Accounting Gap Components by Sector. Stacked bar chart showing median labour imputation, depreciation, and interest by sector.

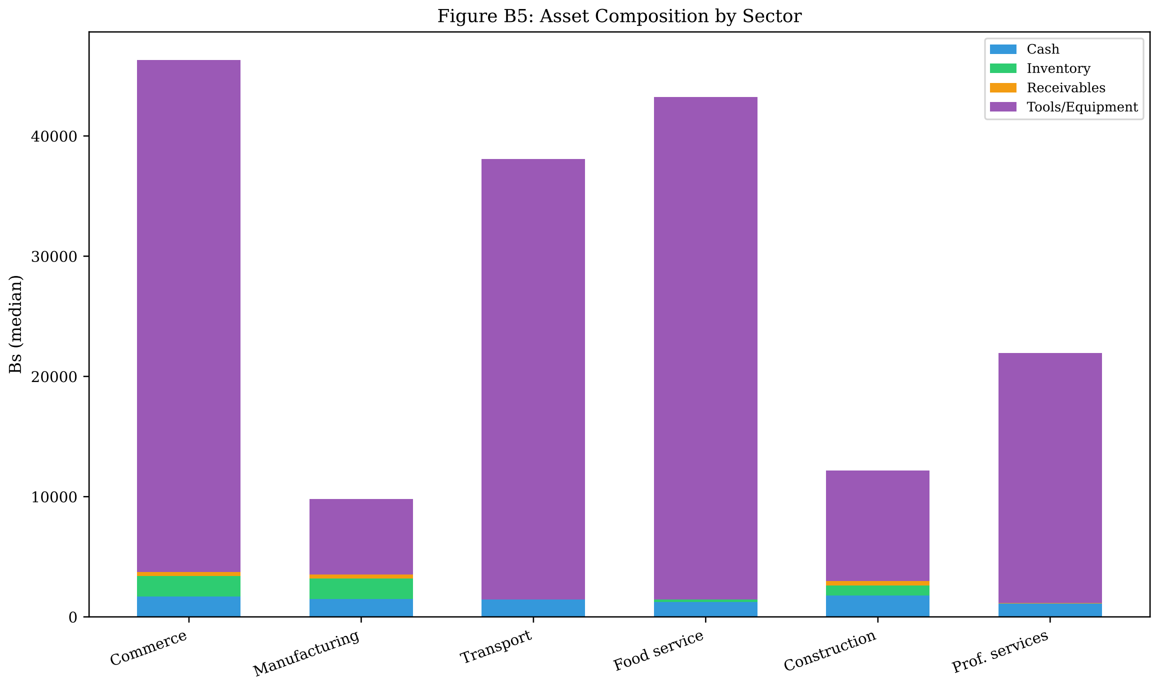

Figure 12: Asset Composition by Sector. Stacked bar chart showing median cash, inventory, receivables, and tools/equipment by sector.

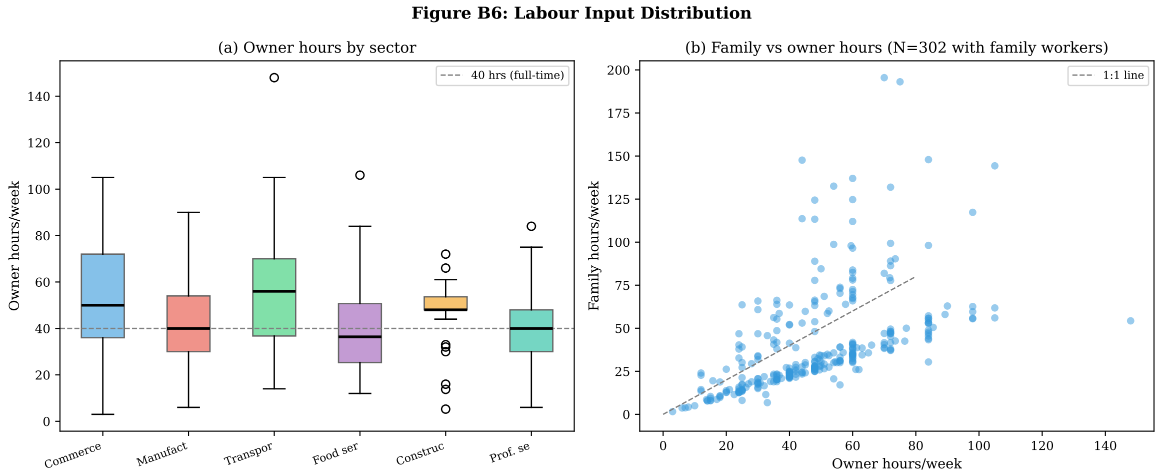

Figure 13: Labour Input Distribution. Panel (a): Owner hours by sector. Panel (b): Family hours vs owner hours for firms with family workers.

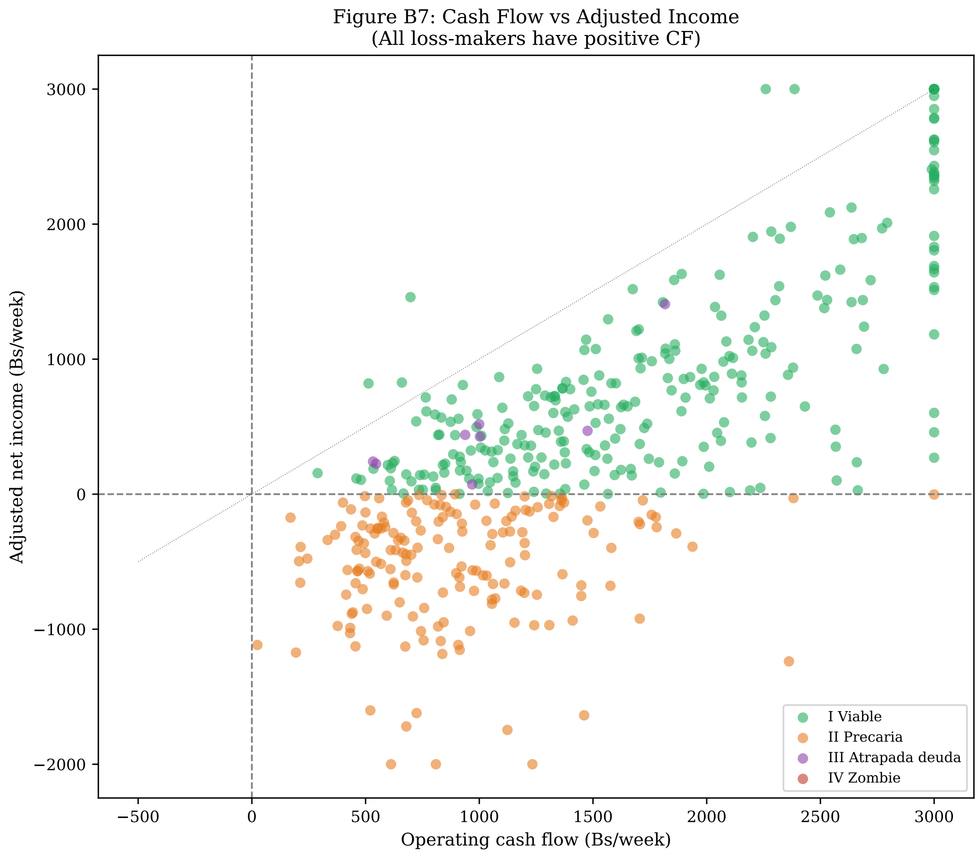

Figure 14: Cash Flow vs Adjusted Income. Each point is one firm, coloured by typology. All loss-making firms (below the horizontal dashed line at zero) have positive operating cash flow, confirming that losses are non-cash. $N = 500$.

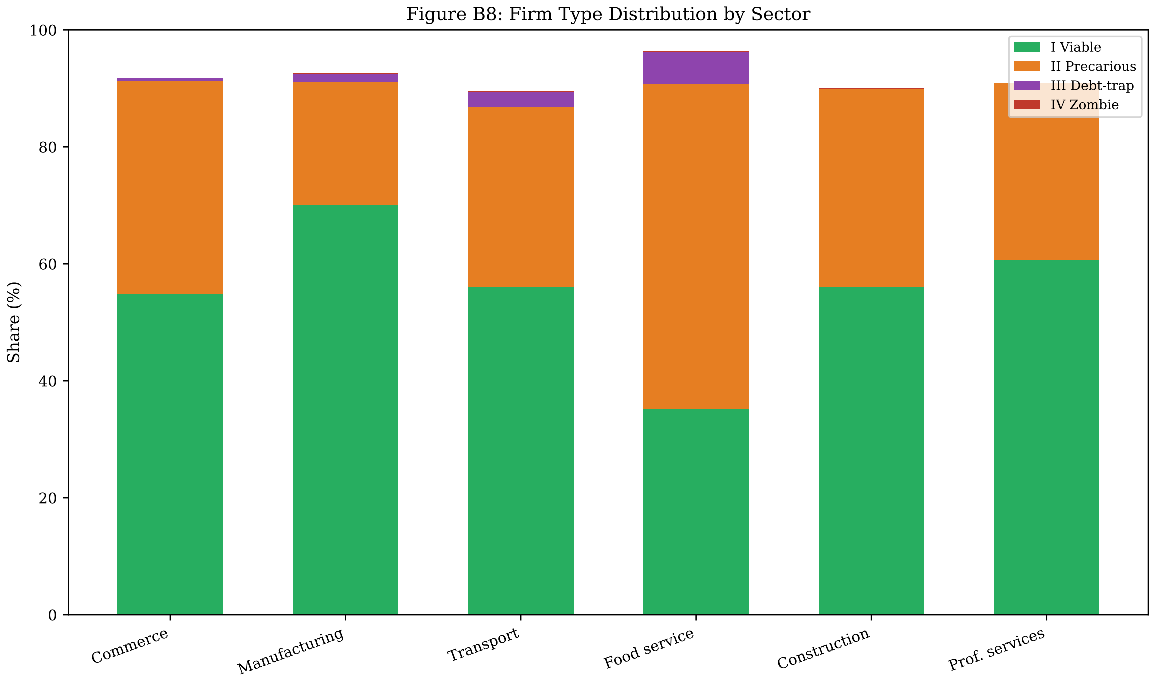

Figure 15: Firm Type Distribution by Sector. Stacked bar chart showing the share of Types I–IV in each sector.

additional materials are available at https://github.com/wernerhl/ledgers-bolivia. A de-identified version of the analysis dataset is available from the author upon request. No personally identifying information is contained in the analysis files.

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